Imbalanced Vehicle Classification

Phase 2 — Midterm Report

Introduction to Artificial Intelligence — Summer 2026

May 2026

1. Implementation Description

1.1 Architecture

We use a ResNet-50 pretrained on ImageNet (torchvision) as our primary backbone. The final fully-connected layer is replaced with a linear layer mapping to 10 output classes. The full network is fine-tuned end-to-end (no frozen layers), which we found necessary to adapt the low-level features to aerial-view imagery.

As a secondary experiment, we also fine-tuned EfficientNet-B0 under identical conditions to serve as a comparison baseline.

1.2 Training Pipeline

The complete training pipeline is implemented in PyTorch and consists of the following steps:

Data Augmentation (training only):

- Random horizontal and vertical flips
- Random rotation (±15°)
- Color jitter: brightness ±0.3, contrast ±0.3, saturation ±0.2
- Resize to 128×128 pixels
- Normalization with ImageNet mean/std

Class Imbalance Strategy (two combined approaches):

- WeightedRandomSampler: each training batch is sampled with probability inversely proportional to class frequency, resulting in roughly balanced batches.
- Weighted CrossEntropyLoss: class weights = 1/count, normalized to sum to num_classes. This penalizes misclassification of minority classes more strongly.

Optimizer and Scheduler:

- AdamW optimizer, learning rate = 1e-4, weight decay = 1e-4
- Cosine Annealing LR scheduler over the full training duration

1.3 Hyperparameters

Parameter	Value
Input resolution	128 × 128
Batch size	64
Learning rate	1e-4
Optimizer	AdamW (weight decay 1e-4)
LR scheduler	Cosine Annealing
Loss function	Weighted CrossEntropyLoss
Epochs run so far	10
Backbone	ResNet-50 (pretrained ImageNet)

2. Preliminary Results

2.1 Quantitative Results

The following table summarizes preliminary results after 10 training epochs on our internal stratified validation split (20% of training data).

Model	Val Accuracy	Macro F1	Minority class F1 (avg)
ResNet-50 — no imbalance handling	0.81	0.43	0.21
ResNet-50 — weighted loss only	0.76	0.61	0.47
ResNet-50 — weighted sampler + weighted loss	0.72	0.68	0.58
EfficientNet-B0 — weighted sampler + weighted loss	0.70	0.65	0.54

Note: overall accuracy decreases when imbalance handling is applied because the model no longer trivially predicts Sedan for all inputs. Macro F1 is the more meaningful metric here, and it improves substantially with our combined strategy.

2.2 Per-class Observations

- Sedan (class 0): recall drops from ~99% to ~72% when imbalance handling is active — the model is less biased toward this class.
- Bus (class 7) and Flatbed truck w/ trailer (class 9): F1 improves from near 0 (baseline) to ~0.52 with combined strategy — the most impactful improvement.
- Pickup truck (class 2) vs Pickup truck w/ trailer (class 8): frequent confusion between these two visually similar classes; still the main error source.

3. Experiment Analysis

3.1 What Worked

- Combining WeightedRandomSampler + weighted loss gave the best Macro F1 (+25 points over baseline). Using either alone was less effective.
- Transfer learning from ImageNet is clearly beneficial — training from scratch for 10 epochs yielded Macro F1 < 0.30.
- Data augmentation (flips, rotation, color jitter) slightly reduces overfitting; validation loss was more stable compared to no augmentation.
- Cosine Annealing scheduler led to smoother convergence compared to a fixed learning rate, which showed instability after epoch 6.

3.2 What Did Not Work / Challenges

- Weighted loss alone (without balanced sampler): the model still over-predicted Sedan in early epochs because most batches were still Sedan-dominated.
- Aggressive augmentation on all classes equally: adding strong random crops caused the model to fail on small-vehicle classes (motorcycle, bus) where cropping removed the vehicle entirely. We reverted to moderate augmentation.
- Higher learning rate (1e-3): caused unstable training loss and poor generalization. Reduced to 1e-4.
- Input resolution 64×64: too small for fine-grained distinctions (e.g. trailer detection). Increased to 128×128 for better detail retention.

3.3 Changes from Initial Plan

- Initial plan proposed 80/20 stratified split → implemented and confirmed working.
- EfficientNet-B0 was tested as planned; ResNet-50 currently performs better (Macro F1: 0.68 vs 0.65), so ResNet-50 remains primary.
- We did not yet implement post-processing (e.g. temperature scaling for calibration) — planned for Phase 3.

3.4 Planned Next Steps

- Train for more epochs (target: 30–50) and select best checkpoint by Macro F1.
- Experiment with larger input resolution (224×224) to capture finer features.
- Try focal loss as an alternative to weighted cross-entropy.
- Analyse the confusion matrix more carefully to apply targeted augmentation for the most confused class pairs.

4. References

[1] S. Low et al., "Multi-modal Aerial View Object Classification Challenge Results - PBVS 2022," CVPRW 2022.

[2] https://codalab.lisn.upsaclay.fr/competitions/1392

[3] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," Journal of Big Data, 2019.

[4] https://towardsai.net/p/l/multi-class-model-evaluation-with-confusion-matrix-and-classification-report